Fields and Waves I

Lecture 5

Power on Lossless Transmission Lines & Lossy Transmission Lines

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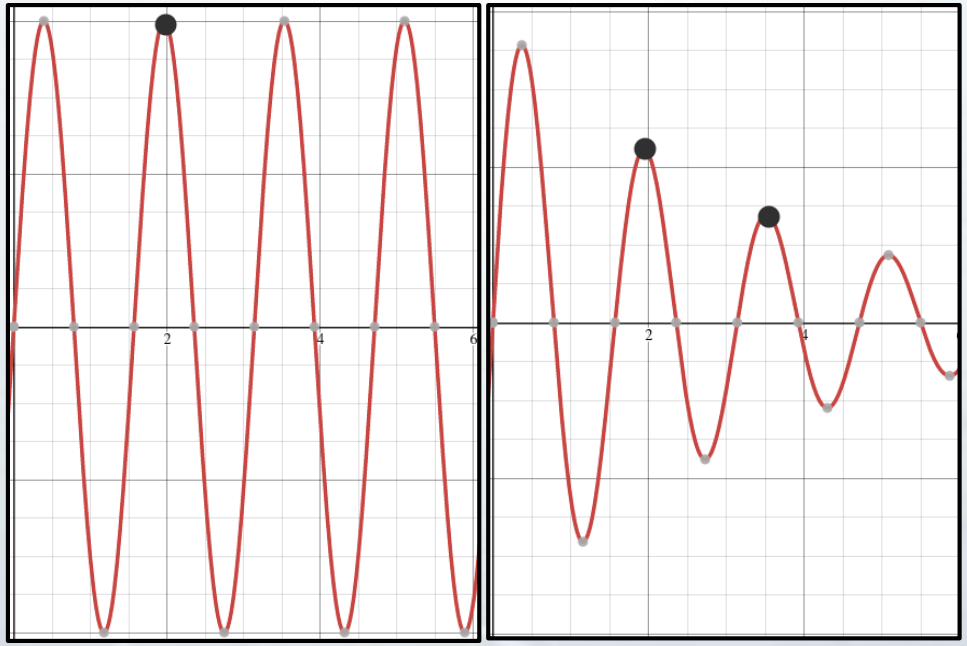
These slides were prepared through the work of the following people:

- Kenneth A. Connor – ECSE Department, Rensselaer Polytechnic Institute, Troy, NY (Emeritus)
- J. Darryl Michael – GE Global Research Center, Niskayuna, NY
- Thomas P. Crowley – National Institute of Standards and Technology, Boulder, CO
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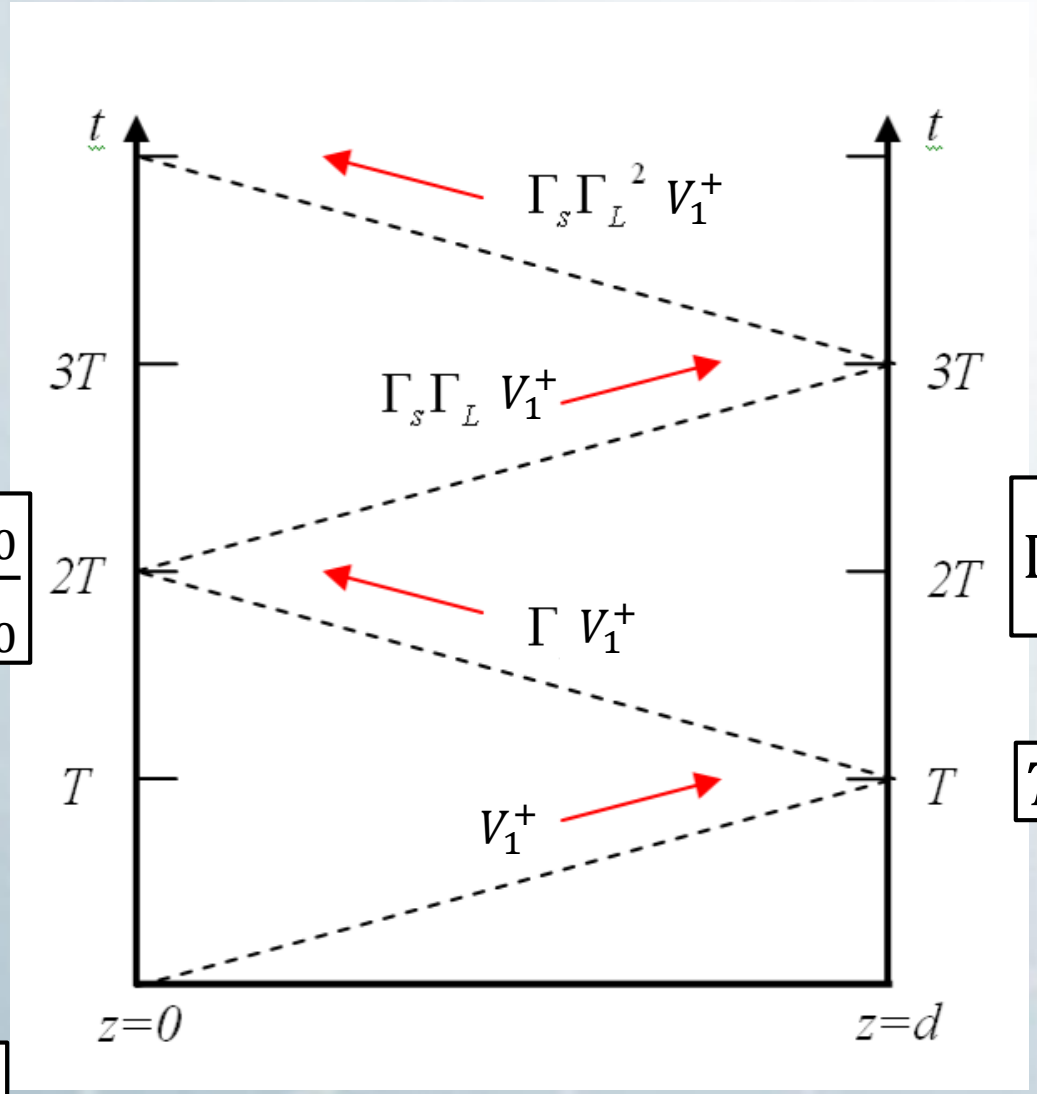
Materials from other sources are referenced where they are used.
Those listed as Ulaby are figures from Ulaby's textbook.

Overview

- Power Analysis on Lossless Transmission Lines
- Lossy Transmission Lines
- Low-Loss Transmission Lines
- Distortionless Transmission Lines



Review



$$\Gamma_g = \frac{Z_g - Z_0}{Z_g + Z_0}$$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$T = d/u$$

Source side

Load side

$$V_1^+ = V_g \frac{Z_0}{Z_g + Z_0}$$

Review

Some things in common between transient and steady state analysis:

- Reflection at either end of the line is governed by the reflection coefficient.
- Both forward- and backward-traveling waves can be present in general.
- Z_0 is needed to understand the relationship between current and voltage on the line.

A major difference between transient and steady state analysis:

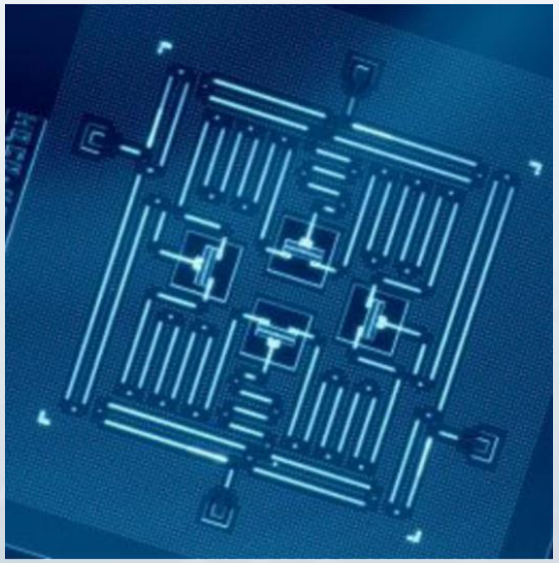
- In transient analysis, there is no backwards-traveling wave **until enough time has passed for a reflection to occur at the end of the line**.

Without forward- and backward-traveling waves on the line at the same time, a lot of our steady-state concepts (standing wave ratio, phase-shifted reflection coefficient) become meaningless.

Power Analysis on Lossless Transmission Lines



https://grist.org/energy/massive-transmission-line-will-send-wind-power-from-wyoming-to-california/



https://en.wikipedia.org/wiki/Transmon

Power Analysis (Lossless T-Line)

- How do we define power in EE?
- Basic definition:

$$P = IV \qquad P = \frac{V^2}{R}$$ for resistive elements

- We can find instantaneous power on the transmission line in the same way:

$$P(d,t) = v(d,t)i^*(d,t)$$

Power Analysis (Lossless T-Line)

$$P(d,t) = v(d,t)i(d,t)$$

- To find **instantaneous incident** power at a location z=-d on the t-line (i.e. we do not consider reflection yet):

$$V(d,t) = \mathrm{Re}\{\tilde{V}e^{j\omega t}\} = \mathrm{Re}\{\tilde{V}^{+}e^{j\omega t}\} + \mathrm{Re}\{\tilde{V}^{-}e^{j\omega t}\}$$

Incident Voltage:
$$V_i(d,t) = V_0^+ \cos(\omega t + \beta d) = |V_0^+|e^{j\phi^+}\cos(\omega t + \beta d)$$
$$= |V_0^+|\cos(\omega t + \beta d + \phi^+)$$

Incident Current:
$$I_i(d,t) = \frac{|V_0^+|}{Z_0}\cos(\omega t + \beta d + \phi^+)$$

Incident Power:
$$P_i(d,t) = \frac{|V_0^+|^2}{Z_0}\cos^2(\omega t + \beta d + \phi^+)$$

Power Analysis (Lossless T-Line)

$$P(d,t) = v(d,t)i(d,t)$$

- To find **instantaneous reflected** power at a location z=-d on the t-line (i.e. we do not consider the incident wave):

$$V(d,t) = \mathrm{Re}\{\tilde{V}e^{j\omega t}\} = \mathrm{Re}\{\tilde{V}^{+}e^{j\omega t}\} + \mathrm{Re}\{\tilde{V}^{-}e^{j\omega t}\}$$

Reflected Voltage:
$$V_r(d,t) = V_0^{-}\cos(\omega t - \beta d) = |V_0^{+}|e^{j\phi^{+}}|\Gamma|e^{j\theta_r}\cos(\omega t - \beta d)$$
$$= |V_0^{+}|\cos(\omega t - \beta d + \phi^{+} + \theta_r)$$

Reflected Current:
$$I_r(d,t) = -\frac{|V_0^{+}||\Gamma|}{Z_0}\cos(\omega t - \beta d + \phi^{+} + \theta_r)$$

Reflected Power:
$$P_r(d,t) = -\frac{|V_0^{+}|^2|\Gamma|^2}{Z_0}\cos^2(\omega t - \beta d + \phi^{+} + \theta_r)$$

↘ convention to tell us it's moving opposite to the incident Power

Power Analysis (Lossless T-Line)

- To find **time-averaged incident power**:

$$P_{av}^{i} = \frac{1}{T}\int_0^T P^i(d,t)dt = \frac{\omega}{2\pi}\int_0^{2\pi/\omega} P^i(d,t)dt$$

$$= \frac{\omega}{2\pi}\int_0^{2\pi/\omega} \frac{|V_0^+|^2}{Z_0} cos^2(\omega t + \beta d + \phi^+)dt$$

$$P_{av}^{i} = \frac{\omega}{2\pi}\frac{\pi}{\omega}\frac{|V_0^+|^2}{Z_0} = \boxed{\frac{|V_0^+|^2}{2Z_0}}$$

$V_{rms} = \frac{V_{peak}}{\sqrt{2}}$

$I_{rms} = \frac{I_{peak}}{\sqrt{2}}$

Power Analysis (Lossless T-Line)

- To find **time-averaged reflected power**:

$$P_{av}^{r} = \frac{1}{T}\int_0^T P^r(d,t)dt = \frac{\omega}{2\pi}\int_0^{2\pi/\omega} P^r(d,t)dt$$

$$= \frac{\omega}{2\pi}\int_0^{2\pi/\omega} -|\Gamma_L|^2\frac{|V_0^+|^2}{Z_0}cos^2(\omega t + \beta d + \phi^+ + \theta^r)dt$$

$$P_{av}^{r} = -|\Gamma|^2\frac{\omega}{2\pi}\frac{\pi}{\omega}\frac{|V_0^+|^2}{Z_0} = \boxed{-|\Gamma_L|^2\frac{|V_0^+|^2}{2Z_0}} = -|\Gamma_L|^2 P_{av}^{i}$$

Power Analysis (Lossless T-Line)

or time-averaged net power

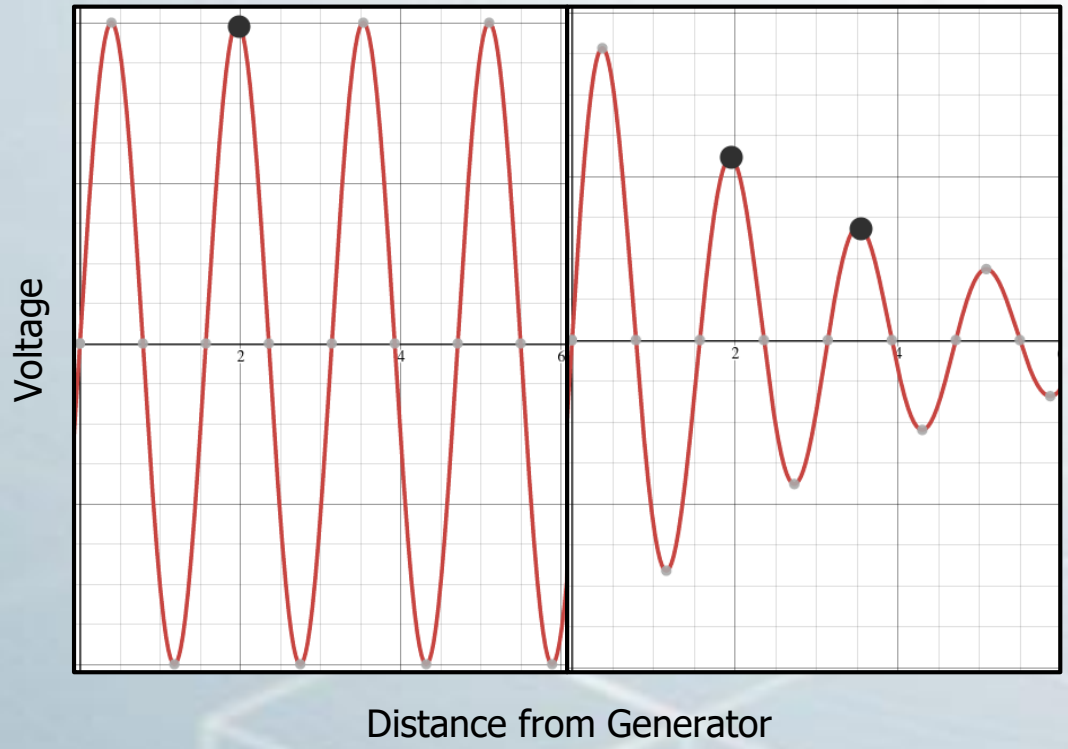
- The **time-average power transmitted to the load**:

$$P_{av} = P_{av}^{i} + P_{av}^{r} = \frac{|V_0^+|^2}{2Z_0}(1 - |\Gamma_L|^2)$$

P_{av}^{r} includes the "−" sign already

Do Lecture 5 Exercise on Gradescope. You may work in groups of up to 4.

Lossy Transmission Lines



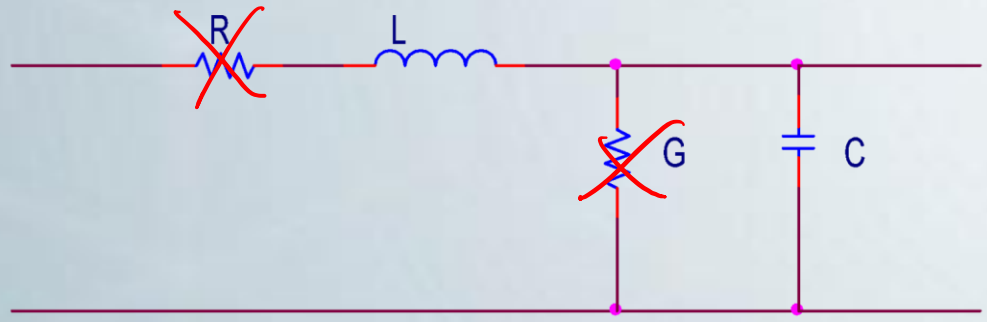
Lossy T-Lines

- In our discussion of transmission lines so far, we have assumed that the lines are lossless – capacitors and inductors exchange energy, but they don't *dissipate* it
- How much attenuation (voltage loss) did you see on the long spool of cable in studio session 2?
- Real transmission lines are never lossless. So how do we generalize what we've learned to "lossy" lines?
- What is missing from our lumped element model?

resistance + conductance

Lossy T-Lines

- Lossless Model of t-line has no R or G (R'=G'=0):



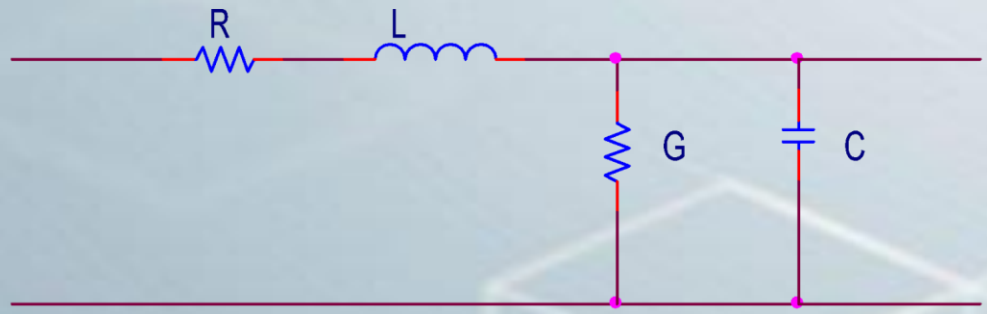
Loss effects due to Resistances:

R - resistance of conductors

G – conductance of insulators

- both are ideally small

- Lossy Model of t-line:

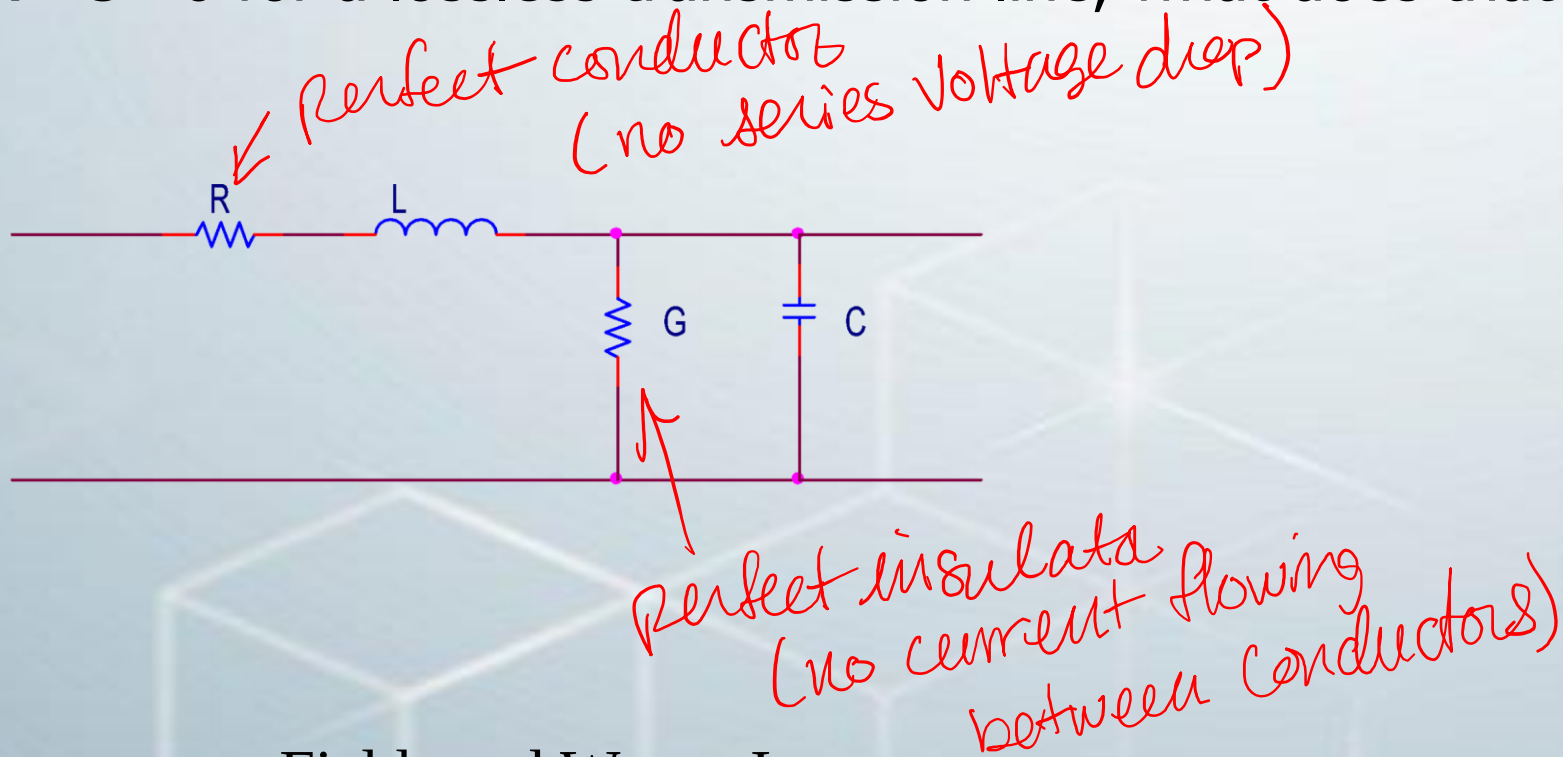


Lossy T-Lines

- What is the difference between impedance and admittance?

$$Z = 1/Y \Rightarrow Y = 1/Z$$

- When we say that R'=G'=0 for a lossless transmission line, what does that physically mean?



Lossy T-Lines

Lossy Telegrapher Equations: Phasor Domain

$$-\frac{d\tilde{V}(z)}{dz} = (r + j\omega l)\tilde{I}(z)$$

$$-\frac{d\tilde{I}(z)}{dz} = (g + j\omega c)\tilde{V}(z)$$

Lossy T-Lines

Lossy Telegrapher Equations: Phasor Domain

$$\frac{d^2\tilde{V}(z)}{dz^2} - (r + j\omega l)(g + j\omega c)\tilde{V}(z) = 0$$

$$\frac{d^2\tilde{I}(z)}{dz^2} - (r + j\omega l)(g + j\omega c)\tilde{I}(z) = 0$$

Alternatively:

$$\frac{d^2\tilde{V}(z)}{dz^2} - \gamma^2\tilde{V}(z) = 0$$

$$\frac{d^2\tilde{I}(z)}{dz^2} - \gamma^2\tilde{I}(z) = 0$$

where $\gamma = \sqrt{(r + j\omega l)(g + j\omega c)}$

← propagation constant → now complex

Lossy T-Lines

Lossy Telegrapher Equations: Phasor Domain

$$\gamma = \sqrt{(r + j\omega l)(g + j\omega c)}$$

$$\gamma = \alpha + j\beta$$

→ same as in lossless lines → propagating wave

$$\alpha = \mathfrak{Re}\left\{\sqrt{(r + j\omega l)(g + j\omega c)}\right\}$$

$$\beta = \mathfrak{Im}\left\{\sqrt{(r + j\omega l)(g + j\omega c)}\right\}$$

For a lossless line, α=0 because r=g=0.

Lossy T-Lines

Lossy Telegrapher Equations: Phasor Domain

$$\tilde{V}(z) = V_0^+ e^{-\gamma z} + V_0^- e^{+\gamma z} = V_0^+ (e^{-\gamma z} + \Gamma_L e^{+\gamma z})$$

$$\tilde{I}(z) = \frac{V_0^+}{Z_0} e^{-\gamma z} - \frac{V_0^-}{Z_0} e^{+\gamma z} = \frac{V_0^+}{Z_0} (e^{-\gamma z} - \Gamma_L e^{+\gamma z})$$

where

$$\gamma = \alpha + j\beta$$

Lossy T-Lines

- For lossless systems: $\beta = \omega\sqrt{lc} = \dfrac{\omega}{u}$
- For lossy systems: $\gamma = \alpha + j\beta = \sqrt{(r + j\omega l)(g + j\omega c)}$

- The phasors now have the factor:

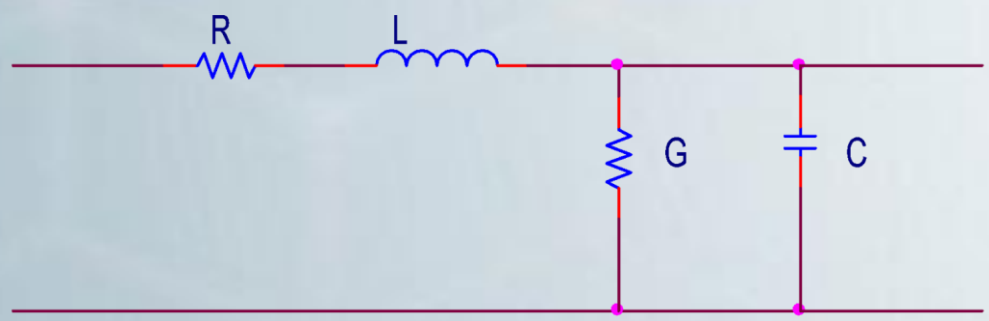
$$e^{-\gamma z} = e^{-\alpha z} \cdot e^{-j\beta z}$$

sinusoidal (lossless)

exponential decay (attenuation) → lossy behavior

Lossy T-Lines

- For a lossless system, Z_0 represents



$$\frac{V_0^+}{I_0^+} = -\frac{V_0^-}{I_0^-} \rightarrow Z_0 = \sqrt{\frac{j\omega l}{j\omega c}} = \sqrt{\frac{l}{c}}$$

- To get the result for **lossy t-lines**:

Replace ***jωl*** with ***r + jωl***

Replace ***jωc*** with ***g + jωc***

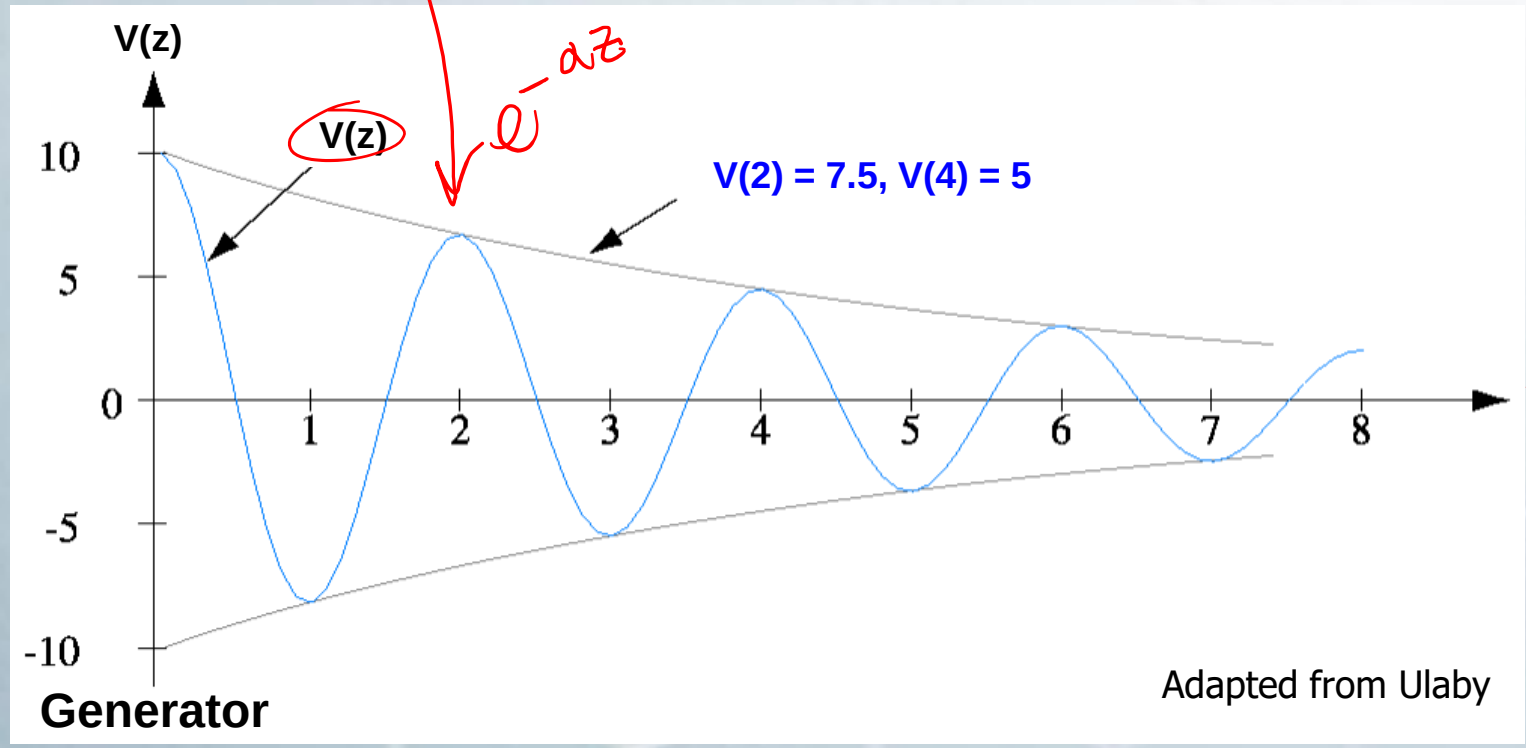
Characteristic Impedance

$$Z_0 = \sqrt{\frac{r + j\omega l}{g + j\omega c}}$$

now complex and introduces a phase shift between V_0^+ and I_0^+

Lossy T-Lines

- The attenuation factor α is due to ohmic losses on the line.
- Looking at a plot like the one below, how do you find α and β?



Lossy T-Lines

- General expression for voltage on the t-line

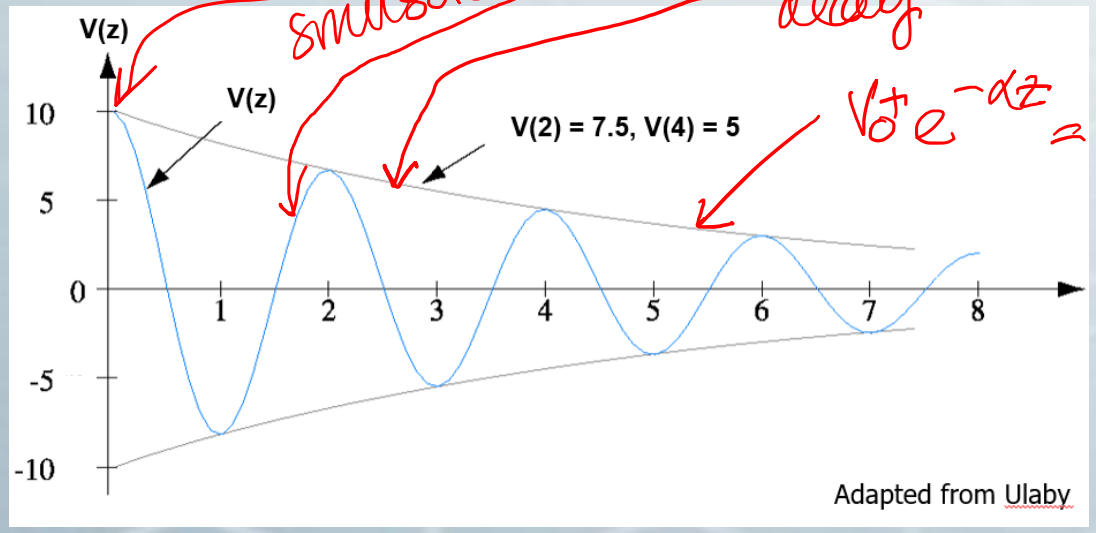
$$\tilde{V}(z) = V_0^+(e^{-\gamma z} + \Gamma_L e^{+\gamma z})$$

- Allowing γ to be complex now introduces a **decaying factor** to the wave

$$e^{-\gamma z} = e^{-\alpha z} \cdot e^{-j\beta z}$$

- The case below shows the effect of this decaying factor on a sinusoid

$$\tilde{V}(z) = V_0^+ \cos(\beta z) e^{-\alpha z}$$



Lossy T-Lines: Solving for α

- Choose 2 convenient points on the voltage envelope

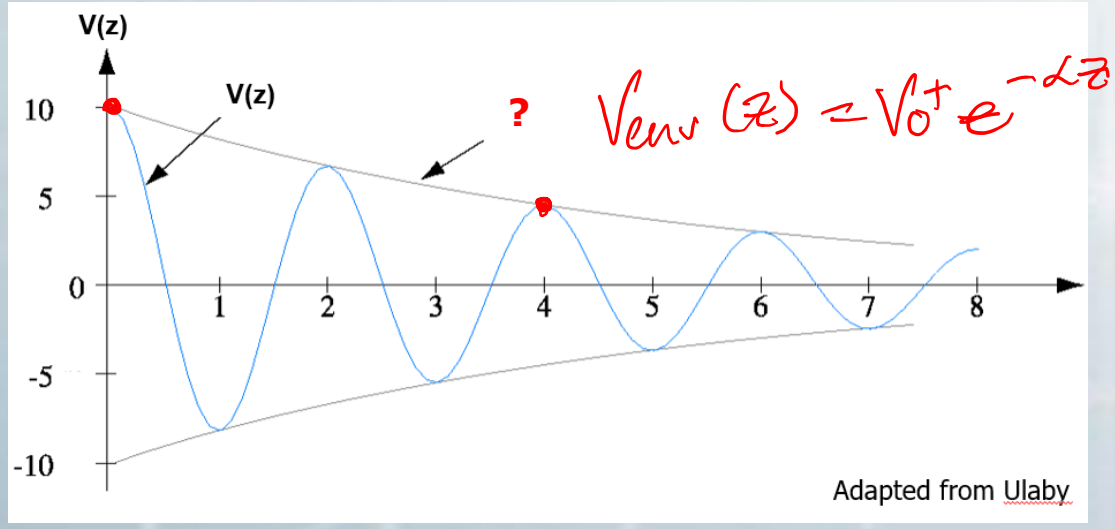
$$\begin{cases} V(0) = 10V \\ V(4m) = 5V \end{cases} \Rightarrow \quad V(0m) = 10V = V_0^+ e^{-\alpha \cdot 0m} = V_0^+$$

$$V(4m) = 5V = 10V\, e^{-\alpha \cdot 4m} \rightarrow 0.5 = e^{-\alpha \cdot 4m} \rightarrow$$

- Plug in chosen values and solve for α

$$\alpha = -\frac{1}{4m}\ln(0.5) = 0.173\ Np/m$$

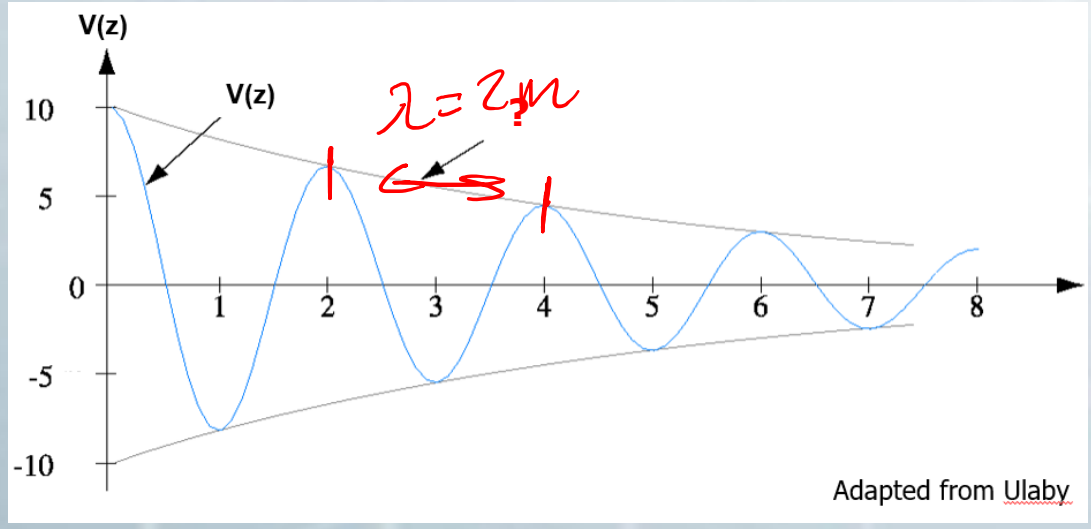
"Nepus"



$$V_{env}(z) = V_0^+ e^{-\alpha z}$$

Lossy T-Lines: Solving for β

$$\beta = \frac{2\pi}{\lambda} = \frac{2\pi}{2m} = \pi \text{ rad/m}$$



Low-Loss Transmission Lines

Low-Loss Lines

- Although real transmission lines are not lossless, in practice the loss of a well-engineered t-line is quite low.

- **Example**: Assume the following:
  - $f = 1\,\mathrm{MHz}$
  - standard RG58 coaxial cable parameters (the ones in lab)
  - r per unit length of 0.1 Ohm per meter

- The wave is seen to attenuate markedly in *2000 meters.*
- How do we model low-loss transmission lines, then?

Low-Loss Lines

- We can derive an expression for Z_0 in the case of a *low-loss transmission line*

- Using the Taylor Series $\sqrt{1+x} \approx 1 + \frac{x}{2}$ for $x \sim 0$.

r is small
↓ g is small

$$Z_0 = \sqrt{\frac{R+j\omega L}{G+j\omega C}} \xrightarrow{\text{assume } G \sim 0} \sim \sqrt{\frac{R+j\omega L}{j\omega C}} = \sqrt{\frac{j\omega L}{j\omega C}}\sqrt{1+\frac{R}{j\omega L}}$$

assume G ~ 0 good assumption

low-loss condition: $R \ll j\omega L$

$$\text{TSE} \sim \sqrt{\frac{L}{C}}\left((1+x)^{1/2}\Big|_{x=0} + \frac{1}{2}(1+x)^{-1/2}\Big|_{x=0} + \dots \right) = \sqrt{\frac{L}{C}}\left(1+\frac{x}{2}\right)$$

$$= \sqrt{\frac{L}{C}}\left(1 - j\frac{R}{2\omega L}\right) \sim \sqrt{\frac{L}{C}} = Z_0$$

Z_0 is approx. the same as in the lossless case

neglect the phase shift (small)

Low-Loss Lines

$$Z_0 = \sqrt{\frac{l}{c}}\left(1 - j\frac{r}{2\omega l}\right)$$

- Note that in the low-loss case, the following will be true:

$$|Z_0| \approx |\sqrt{\frac{l}{c}}(1 - j\frac{r}{2\omega l})| \approx \sqrt{\frac{l}{c}}$$

- For low-loss lines, Z_0 is then the same as for lossless lines.
- This equation is **sufficient when you do not care about the phase of Z_0**, since this angle is small and can often be ignored in practice.

Low-Loss Lines

- What happens to the propagation and attenuation coefficients when the line is low-loss?

$$\gamma = \alpha + j\beta \approx j\omega\sqrt{lc}\left(1 - j\frac{r}{2\omega l}\right) = j\omega\sqrt{\ell c} + \frac{r}{2}\sqrt{\frac{c}{\ell}}$$

- The propagation and attenuation constants become

$$\alpha = \frac{r}{2}\sqrt{\frac{c}{\ell}} = \frac{r}{2Z_0}$$

$$\beta = \omega\sqrt{\ell c} = \omega/u \leftarrow \text{same as lossless case}$$

Low-Loss Lines

- Assume the following: $f = 1\ \text{MHz}$ & standard RG58 cable parameters & r per unit length of 0.1 Ohm per meter, the wave is seen to attenuate markedly in *2000 meters.* Calculate α and β for this low-loss line. Vf = 0.66

$$\alpha = \frac{r}{2Z_0} = \frac{0.1\,\Omega/m}{2 \cdot 50\,\Omega} = 0.001\ m^{-1}$$

$$\beta = \omega/u = \frac{2\pi \cdot 10^6\,Hz}{0.66 \cdot 3\times10^8\,m/s} = 0.032\ m^{-1}$$

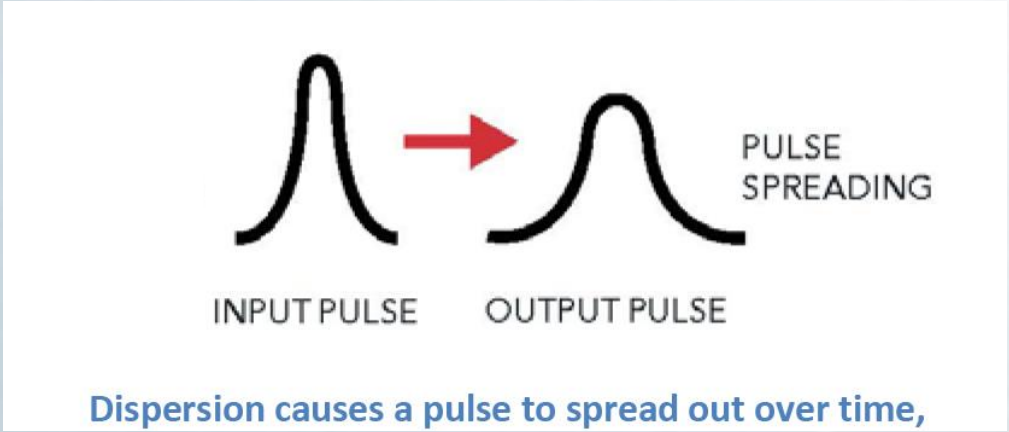
1V

$$\tilde{V}(z) = V_0^+ e^{-\alpha z} = 1V \cdot e^{-0.001 m^{-1} \cdot 2000m} = 0.135\ V$$

86.5% attenuation

Other T-Line Parameters

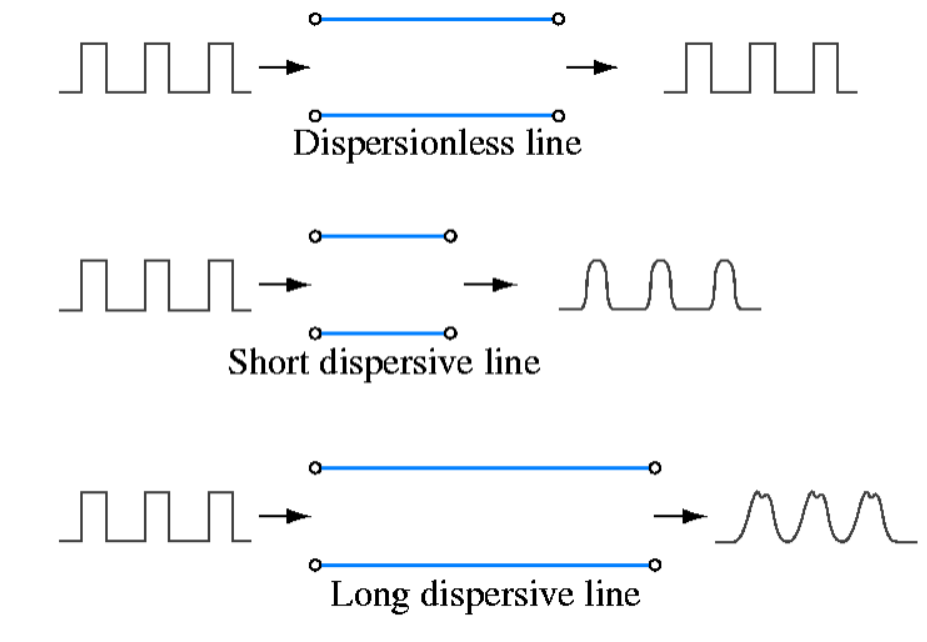
- Aside from attenuation (loss of amplitude), what other problems could you have with your signal after running it through a very long transmission line with losses?



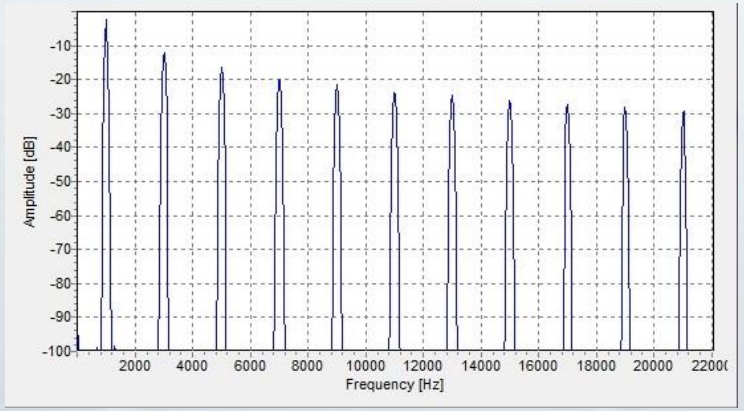
Dispersion causes a pulse to spread out over time,

Other T-Line Parameters

Dispersion: A dispersive transmission line will have frequency-dependent impedance behavior, leading to distortion of signals. (Keep in mind that a square pulse is composed of a series of harmonic frequencies)



Square wave frequency spectrum



https://upload.wikimedia.org/wikipedia/commons/b/b5/Spectrum_square_oscillation.jpg

Other T-Line Parameters

- Note that the propagation constant varies with frequency

$$\gamma = \alpha + j\beta = \sqrt{(r + j\omega l)(g + j\omega c)}$$

- Z_o is also frequency dependent and not purely resistive

$$Z_0 = \sqrt{\frac{r + j\omega l}{g + j\omega c}}$$

- Can we fix this somehow so that our signals are not distorted?

Distortionless Transmission Lines

Distortionless Lines

- Distortion in a transmission line limits its useful length.
- Attenuation can be addressed by adding amplification, however, distorted signals cannot generally be undistorted, so a method needed to be found to eliminate it.
- Remarkably, lines can be made distortionless by adding loss. That is, we can trade additional attenuation for clarity of signal.

Distortionless Lines

- For practical lines, the conductance per unit length g is negligible. What can we do to eliminate the frequency dependence of Z_0?

$$Z_0 = \sqrt{\frac{r+j\omega\ell}{g+j\omega c}} = \sqrt{\frac{\ell}{c}}\sqrt{\frac{r/\ell+j\omega}{g/c+j\omega}} \Rightarrow r/\ell + j\omega = g/c + j\omega$$

set to 1 and solve

$$\boxed{r/\ell = g/c}$$

This is called the **Heaviside condition** and it can be achieved with periodic lumped shunt resistors.

Distortionless Lines

- For this combination of parameters

$$\gamma = \alpha + j\beta = \sqrt{(r + j\omega l)\left(\frac{rc}{l} + j\omega c\right)} = \sqrt{\frac{c}{l}}\left(r + j\omega l\right)$$

when using Heaviside condition:

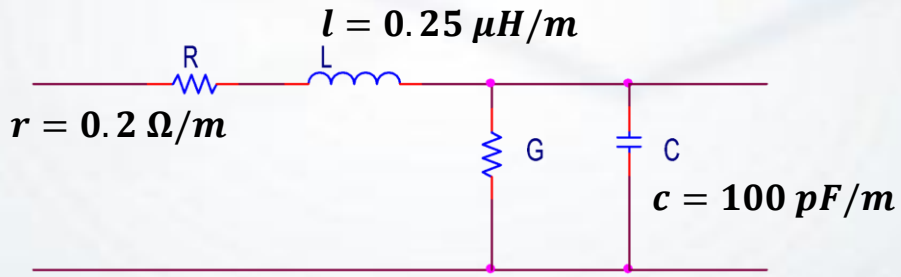
$$\alpha = r\sqrt{\frac{c}{l}}$$

$$\beta = \omega\sqrt{lc} = \frac{\omega}{u}$$

$$Z_0 = \sqrt{\frac{l}{c}}$$

Distortionless Lines

- Consider a "bad RG-58" cable, which has
  - L = 50m,
  - l = 0.25μH/m,
  - c = 100pF/m
  - r = 0.2Ω/m.



- What g is required to make it distortionless?

$$r/l = g/c \rightarrow g = \frac{rc}{l} = \frac{0.2\Omega/m \cdot 100\times10^{-12}\ F/m}{0.25\times10^{-6}\ H/m} = 8\times10^{-5}\ S/m = 80\mu S/m$$

To figure out how much resistance / length: $G = g \cdot L = 80\mu S/m \cdot 50m = 4\times10^{-4}\ S$

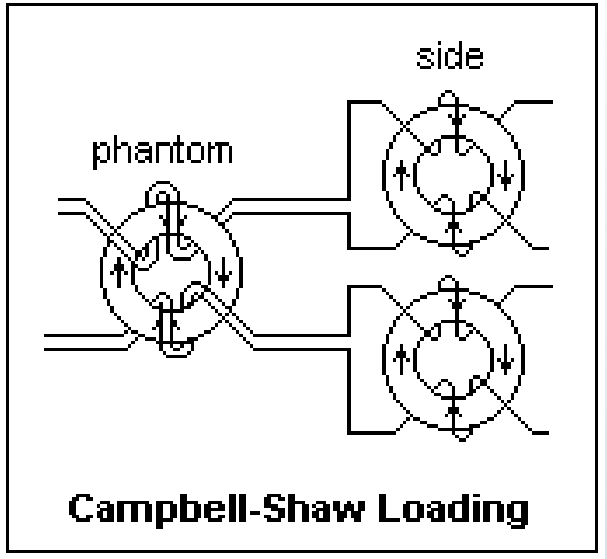
total resistance

$$R = 1/G = 2500\Omega \rightarrow \frac{R}{L} = \frac{2500\Omega}{50m} = 50\Omega$$ shunt resistance every meter

Distortionless Lines

- In the early days of telephony, Heaviside proposed making lines distortionless. This was done by adding inductance rather than conductance since the losses were not increased significantly.

$$\frac{r}{l} = \frac{g}{c}$$



http://www.du.edu/~jcalvert/tech/cable.htm

Distortionless Lines

- Adding these components made it possible for phone calls to go from NY to Chicago.
- Then in the 1850s an even more impressive engineering feat was achieved: the first transatlantic undersea cable. This would not have been possible without knowledge of the Heaviside Condition.

Distortionless Lines



Distortionless Lines

Modern submarine cable with repeaters – counteracts attenuation



prog.world

Wrap-Up

- Studio Session #3 **Wednesday**